

Supplementary file: Dual-channel graph learning of similarity- and complementarity-driven interactions in protein–protein interaction networks

September 4, 2025

1 Detailed performance of DMG-PPI and benchmark methods

Table S1: Performance on SHS27K with BFS scheme

method	Micro-F1	AUPRC	LRAP	NDGC	AUROC	z-score
DMG-PPI	0.81 ± 0.00	0.70 ± 0.01	0.73 ± 0.01	0.89 ± 0.00	0.86 ± 0.00	1.56 ± -1.14
MAPE-PPI	0.71 ± 0.01	0.62 ± 0.01	0.64 ± 0.01	0.83 ± 0.01	0.78 ± 0.01	0.32 ± -0.42
BaPPI	0.76 ± 0.01	0.63 ± 0.01	0.66 ± 0.01	0.87 ± 0.01	0.83 ± 0.01	0.80 ± -0.40
HIGH-PPI	0.70 ± 0.00	0.58 ± 0.00	0.61 ± 0.01	0.83 ± 0.01	0.78 ± 0.00	0.04 ± -0.64
LDMGNN	0.69 ± 0.02	0.59 ± 0.01	0.63 ± 0.01	0.83 ± 0.01	0.76 ± 0.01	0.09 ± -0.22
AFTGAN	0.69 ± 0.01	0.57 ± 0.02	0.60 ± 0.01	0.83 ± 0.01	0.77 ± 0.00	-0.04 ± 0.33
GNN-PPI	0.62 ± 0.02	0.51 ± 0.02	0.56 ± 0.02	0.81 ± 0.01	0.72 ± 0.01	-0.74 ± 0.93
PIPR	0.48 ± 0.05	0.43 ± 0.02	0.50 ± 0.01	0.75 ± 0.01	0.63 ± 0.02	-2.03 ± 1.57

Table S2: Performance on SHS27K with DFS scheme

method	Micro-F1	AUPRC	LRAP	NDGC	AUROC	z-score
DMG-PPI	0.83 ± 0.01	0.73 ± 0.01	0.76 ± 0.00	0.89 ± 0.01	0.88 ± 0.01	1.65 ± -0.05
MAPE-PPI	0.73 ± 0.00	0.62 ± 0.00	0.65 ± 0.00	0.81 ± 0.00	0.80 ± 0.00	0.18 ± -1.11
BaPPI	0.77 ± 0.01	0.63 ± 0.01	0.67 ± 0.01	0.85 ± 0.01	0.85 ± 0.00	0.69 ± -0.47
HIGH-PPI	0.73 ± 0.00	0.62 ± 0.01	0.64 ± 0.02	0.83 ± 0.01	0.80 ± 0.00	0.21 ± -0.08
LDMGNN	0.70 ± 0.01	0.59 ± 0.01	0.63 ± 0.01	0.80 ± 0.00	0.78 ± 0.01	-0.17 ± -0.31
AFTGAN	0.71 ± 0.01	0.60 ± 0.01	0.63 ± 0.02	0.81 ± 0.02	0.79 ± 0.01	-0.03 ± 1.28
GNN-PPI	0.67 ± 0.01	0.55 ± 0.01	0.62 ± 0.01	0.80 ± 0.01	0.76 ± 0.00	-0.44 ± -0.31
PIPR	0.55 ± 0.02	0.43 ± 0.01	0.49 ± 0.01	0.73 ± 0.01	0.67 ± 0.01	-2.10 ± 1.04

Table S3: Performance on SHS148K with BFS scheme

method	Micro-F1	AUPRC	LRAP	NDGC	AUROC	z-score
DMG-PPI	0.81 ± 0.01	0.70 ± 0.01	0.73 ± 0.01	0.88 ± 0.01	0.85 ± 0.01	1.32 ± 0.02
MAPE-PPI	0.77 ± 0.00	0.69 ± 0.01	0.70 ± 0.01	0.85 ± 0.01	0.82 ± 0.00	0.75 ± -0.94
BaPPI	0.78 ± 0.02	0.66 ± 0.03	0.68 ± 0.04	0.85 ± 0.02	0.83 ± 0.01	0.68 ± 2.04
HIGH-PPI	0.77 ± 0.00	0.69 ± 0.01	0.70 ± 0.01	0.85 ± 0.01	0.81 ± 0.00	0.76 ± -0.57
LDMGNN	0.62 ± 0.00	0.55 ± 0.02	0.61 ± 0.03	0.79 ± 0.01	0.70 ± 0.01	-1.04 ± 0.26
AFTGAN	0.71 ± 0.01	0.64 ± 0.01	0.66 ± 0.01	0.83 ± 0.00	0.77 ± 0.00	0.09 ± -0.67
GNN-PPI	0.61 ± 0.01	0.55 ± 0.01	0.63 ± 0.01	0.80 ± 0.00	0.70 ± 0.00	-0.98 ± -0.83
PIPR	0.57 ± 0.01	0.52 ± 0.02	0.58 ± 0.02	0.78 ± 0.01	0.67 ± 0.01	-1.57 ± 0.70

Table S4: Performance on SHS148K with DFS scheme

method	Micro-F1	AUPRC	LRAP	NDGC	AUROC	z-score
DMG-PPI	0.84 ± 0.00	0.76 ± 0.01	0.78 ± 0.01	0.90 ± 0.00	0.88 ± 0.00	1.13 ± -0.22
MAPE-PPI	0.82 ± 0.01	0.73 ± 0.01	0.74 ± 0.01	0.87 ± 0.00	0.86 ± 0.00	0.60 ± 0.27
BaPPI	0.82 ± 0.01	0.72 ± 0.01	0.75 ± 0.01	0.88 ± 0.00	0.87 ± 0.01	0.76 ± 0.76
HIGH-PPI	0.77 ± 0.00	0.68 ± 0.00	0.71 ± 0.01	0.85 ± 0.01	0.83 ± 0.00	0.03 ± 0.02
LDMGNN	0.77 ± 0.00	0.68 ± 0.01	0.71 ± 0.00	0.85 ± 0.00	0.82 ± 0.00	0.03 ± -0.10
AFTGAN	0.79 ± 0.00	0.70 ± 0.01	0.73 ± 0.01	0.86 ± 0.00	0.84 ± 0.00	0.32 ± 0.68
GNN-PPI	0.72 ± 0.00	0.63 ± 0.01	0.68 ± 0.01	0.83 ± 0.00	0.79 ± 0.00	-0.56 ± -0.72
PIPR	0.58 ± 0.00	0.50 ± 0.01	0.58 ± 0.00	0.78 ± 0.00	0.69 ± 0.00	-2.31 ± -0.70

2 Interpretability of DMG-PPI

DMG-PPI identifies hotspot residues within PPIs. Shown below are 11 representative cases, where hotspot residues are highlighted in red. For each case, the left panel displays the ground truth, while the right panel shows the prediction computed by DMG-PPI.



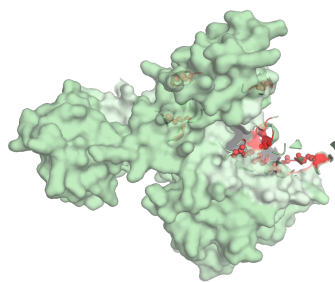
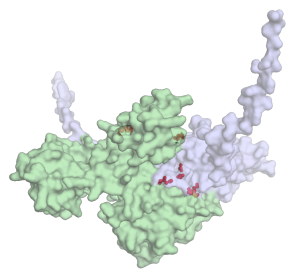
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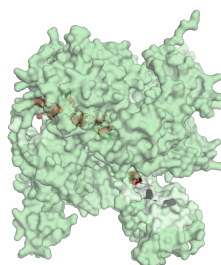
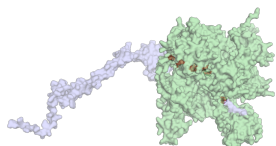
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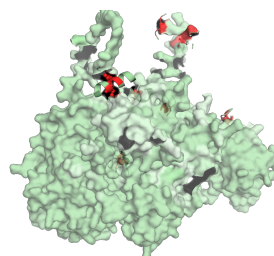
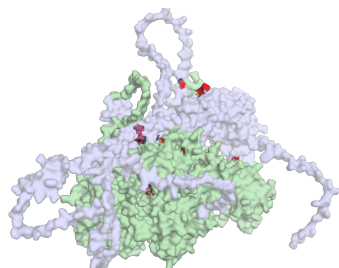
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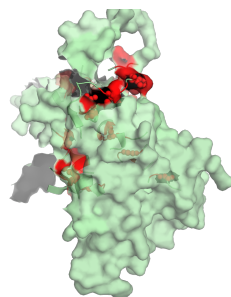
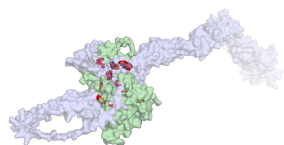
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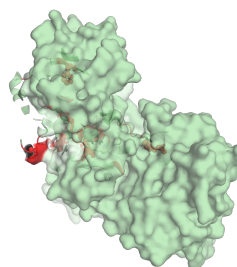
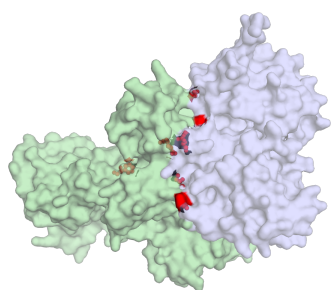
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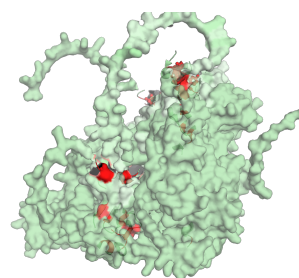
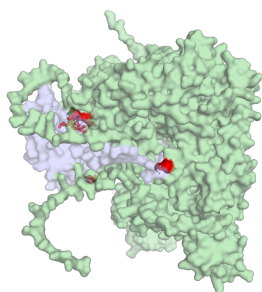
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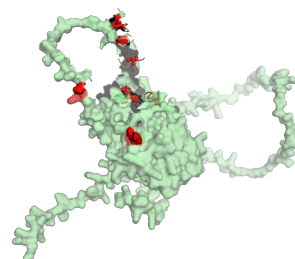
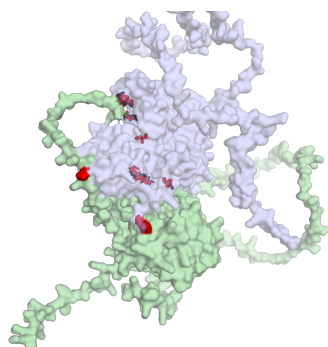
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P53778 and P41240



Q9NP95 and P19174



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